

PV181 Laboratory of security and applied cryptography



Public key crypto Common math operations

Marek Sýs

syso@mail.muni.cz, A404

CRCS

Centre for Research on
Cryptography and Security

You will learn

- How to generate public/private key pair.
- Formats of cryptographic data
 - Base64,
 - ASN1 (PEM, DER)
- Mathematical operations used in public-key cryptography
 - Modular operations
 - Operations with points on Elliptic curve

Public key operations

- Modular operations:
 - Addition
 - Multiplication
 - Exponentiation
 - Inversion (uses Extended Euclid alg.)
- Elliptic curve operations:
 - Point addition
 - Point multiplication (e.g. point doubling $2*P = P+P$)

Modular operations

- The result always in $[0, m-1]$ for modulus m
- Addition, multiplication: $a + b \bmod m$, $a * b \bmod m$
 - In python **$a+b \% m$, $a*b \% m$**
- Exponentiation: $a^e \bmod m$
 - In python **$\text{pow}(a, e, m)$**
 - Computed iteratively with modulo applied to decrease size of intermediate products
 - $a^{11} \bmod m \Rightarrow 11 = 1 + 2 + 8 \Rightarrow a^{11} = a^1 * a^2 * a^8$
 - $a^1, a^2 = a * a \bmod m, a^4 = a^2 * a^2 \bmod m, a^8, a^{16}$
 - $a^{11} \bmod m = ((a^1 * a^2) \bmod m * a^8) \bmod m$

Modular inversion

- Inverse b^{-1} of b modulo m
 - b^{-1} such that $b^{-1} * b = b * b^{-1} = 1 \bmod m$
 - b and m must be coprime!
- Modular “division” using inverse:
$$\frac{a}{b} \bmod m = a * b^{-1} \bmod m$$
 - In python **(a * pow(a, -1, m)) %**
- Examples:
 - $3^{-1} \bmod 7 = 5$ since $3 * 5 \bmod 7 = 1$
 - $4^{-1} \bmod 10$ does not exist ($\gcd(4, 10) = 2$)

RSA

- Public N, e
 - Encryption – modular exponentiation
 - $E(m) = m^e \bmod N$
- Private e, d, p, q, N
 - Decryption – modular exponentiation
 - $D(c) = c^d \bmod N$
- Private
 - Decryption – modular exponentiation using CRT

Elliptic curve cryptography (ECC)

- Defined by parameters
 - a, b, p, G, q
- Groups of points (x, y)
 - $y^2 = x^3 + ax + b \bmod p$
- Operations with points:
 - Addition - complex [wiki](#) $(x_1, y_1) + (x_2, y_2) \neq (x_3, y_3)$
 - Multiplication vs addition – classical relationship
 - E.g. $3 * (x_1, y_1) = (x_1, y_1) + (x_1, y_1) + (x_1, y_1)$
 - Maximal multiplier q i.e.
 - $k * (x_1, y_1) = (k \bmod q) * (x_1, y_1)$